

MS BITHI

AI Security Researcher | Federated Learning & Cybersecurity

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PROFILE SUMMARY

My research journey began with DDoS detection using ensemble machine learning during my undergraduate thesis, inspired by a professor's work in cybersecurity that first captivated me. This led me to specialize in Federated Learning for privacy-preserving security systems, where I have designed non-IID-robust aggregation algorithms for intrusion detection across UAV, IoT, and maritime radar systems, publishing two first-author papers in *Scientific Reports* (Nature Portfolio, Q1). My research has since expanded into emerging directions within Federated Learning, including federated unlearning, feature heterogeneity, and healthcare applications, reflecting my broader interest in adapting Federated Learning and Cybersecurity to diverse real-world domains.

EDUCATION

Bachelor of Science in Computer Science & Engineering

Feb 2020 – May 2024

Prime University, Dhaka, Bangladesh

CGPA: 3.83 / 4.00

(Merit Scholarships from 7th semester onward)

- **Undergraduate Thesis:** Ensemble ML approach for DDoS detection in SDN: Applied RFE feature selection; published at IEEE ICEEICT 2024. DOI: [10.1109/ICEEICT62016.2024.10534483](https://doi.org/10.1109/ICEEICT62016.2024.10534483)

RESEARCH INTERESTS

Cybersecurity & Network Intrusion Detection, Federated & Distributed Learning, Explainable AI (XAI) for Security, LLM/VLM-based Security Automation, Adversarial ML & Model Robustness, Agentic AI for Cyber Defense.

RESEARCH PUBLICATIONS

Journal Papers

- P1. [Scientific Reports 2026] Ms. Bithi, T. Alsubait, A. Ibraheem, M. E. Masud, and M. A. Hossain, "A federated learning-benchmarking framework for privacy-preserving UAV intrusion detection using adaptive aggregation algorithms," *Scientific Reports*, vol. 16, Article 21664, 2026.
DOI: [10.1038/s41598-026-50865-9](https://doi.org/10.1038/s41598-026-50865-9)
- P2. [Scientific Reports 2026] Ms Bithi, M. E. Masud, and M. A. Hossain, "A new adaptive federated learning approach for privacy preserving UAV anomaly detection under non-IID distributions," *Scientific Reports*, vol. 16, Article 8451, 2026.
DOI: [10.1038/s41598-026-38732-z](https://doi.org/10.1038/s41598-026-38732-z)

Conference Paper

- P3. [ICEEICT 2024] Ms. Bithi, M. A. Hossain, M. K. Ahmed, R. Sultana, I. Ahammad, and M. S. Islam, "Enhanced DDoS Detection in Software Defined Networking Using Ensemble-Based Machine Learning," *2024 6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT)*, IEEE, 2024.
DOI: [10.1109/ICEEICT62016.2024.10534483](https://doi.org/10.1109/ICEEICT62016.2024.10534483)

CURRENT RESEARCH (UNDER REVIEW)

- **GEM-FedMAR:** A Privacy Preserving Federated Dual Branch Cross Attention Framework for Maritime Radar Intrusion Detection.
- **Cluster-Aware Federated Aggregation for Heterogeneous Maritime Radar Intrusion Detection under IID and Non-IID Data Distributions.** (*ClusterAvg: novel clustering-based FL aggregation, RadarPWN dataset, F1: 0.9988*)

RESEARCH PROJECTS

Voice Fingerprinting Attack – Network Traffic Analysis

2025 – Present

- Used Wireshark-captured dataset (Kennedy et al., 2019) from Amazon Echo smart speaker (1,000 traffic traces, 100 voice commands).
- Extracted network-level features: upstream/downstream volume, burst patterns, packet timing, trace duration.
- Implemented and benchmarked 4 fingerprinting attack algorithms: LL-Jaccard, LL-NB (Naive Bayes), VNG++ (403-dim burst features), P-SVM (AdaBoost, 405-dim combined features).
- Used Doc2Vec semantic embeddings trained on Quora dataset for semantic similarity of voice commands.

Other Projects:

- MNIST (ANN), Customer Churn (ANN), IoT Attack Detection (TFF), NSL-KDD (SVM, RF), DSDM Simulation, Synthetic Data Generation for FL.

TECHNICAL SKILLS

- **Federated & Distributed ML:** FedAvg, FedAdam, FedProx, FedMedian, ClusterAvg; non-IID data handling, Dirichlet partitioning.
- **Network & Security:** Wireshark (traffic capture & analysis), network feature extraction (upstream/downstream, burst patterns, packet timing), IDS/anomaly detection, SDN, DDoS defense.
- **Machine Learning:** Ensemble Modeling, XGBoost, RF, SVM, RFE Feature Selection.
- **Deep Learning:** CNN, RNN, LSTM, ANN, Dual-Branch CNN, Cross-Attention, GEM, SMOTE, Stratified Splitting.
- **Generative AI & LLMs (In Progress):** Transformer architecture, Self-Attention, Positional Encoding, LLM fine-tuning (LoRA, QLoRA, RLHF), RAG, Diffusion models, Agentic AI systems.
- **Programming & Frameworks:** Python, C/C++, SQL, TensorFlow, PyTorch, Keras, Scikit-learn, Flower, TensorFlow Federated (TFF).
- **Data & Tools:** Zotero, Google Colab, Synthetic Data Generation, Model Evaluation, Doc2Vec, Kaggle, Anaconda, Git, Visual Studio.

PROFESSIONAL EXPERIENCE

Save the Children	Dhaka, Bangladesh
<i>Junior Software Engineer</i>	Nov 2024 – Present
<i>Trainee .NET Developer</i>	Sep 2024 – Nov 2024
• Promoted to Junior Software Engineer after demonstrating technical proficiency in enterprise software development during initial tenure. • Developing and maintaining enterprise application modules using ASP.NET Core MVC, Blazor, and C#. • Contributing to software solution design, code optimization, and database integration for organization-wide tools.	

PRESENTATIONS & ACHIEVEMENTS

- **IEEE Conference Paper Presentation – ICCCNT 2024:** Enhanced DDoS Detection in SDN
- **Champion Poster Presentation – CSE Fest 2024:** DDoS Attack Detection using Feature Selection & ML
- **Thesis Presentation Excellence:** Recognized for technical depth and clarity
- **HackerRank SQL Badge:** Solved all SQL challenges
- **Merit Scholarships:** Awarded from 7th semester onward

LEADERSHIP & EXTRACURRICULAR

General Secretary & Treasurer – Prime University Computer Club	2022 – 2024
• Organized annual CSE festivals & technical seminars (+20% participation). • Conducted training sessions on programming & problem-solving (mentored 50+ peers).	
Campus Ambassador – GUVI	2022 – 2023
Active Volunteer – National Events & Blood Donation Club	

RELEVANT COURSES & LANGUAGE

- **Courses:** Cryptography & Network Security, Computer Networking, Artificial Intelligence, Machine Learning, Discrete Mathematics, Data Structures and Algorithms.
- **Language: English:** Fluent (medium of instruction throughout undergraduate studies); **IELTS:** Scheduled for November 2026; **Bengali:** Native.